Visual Programming

C# (C sharp)

.Net Framework
Visual Studio

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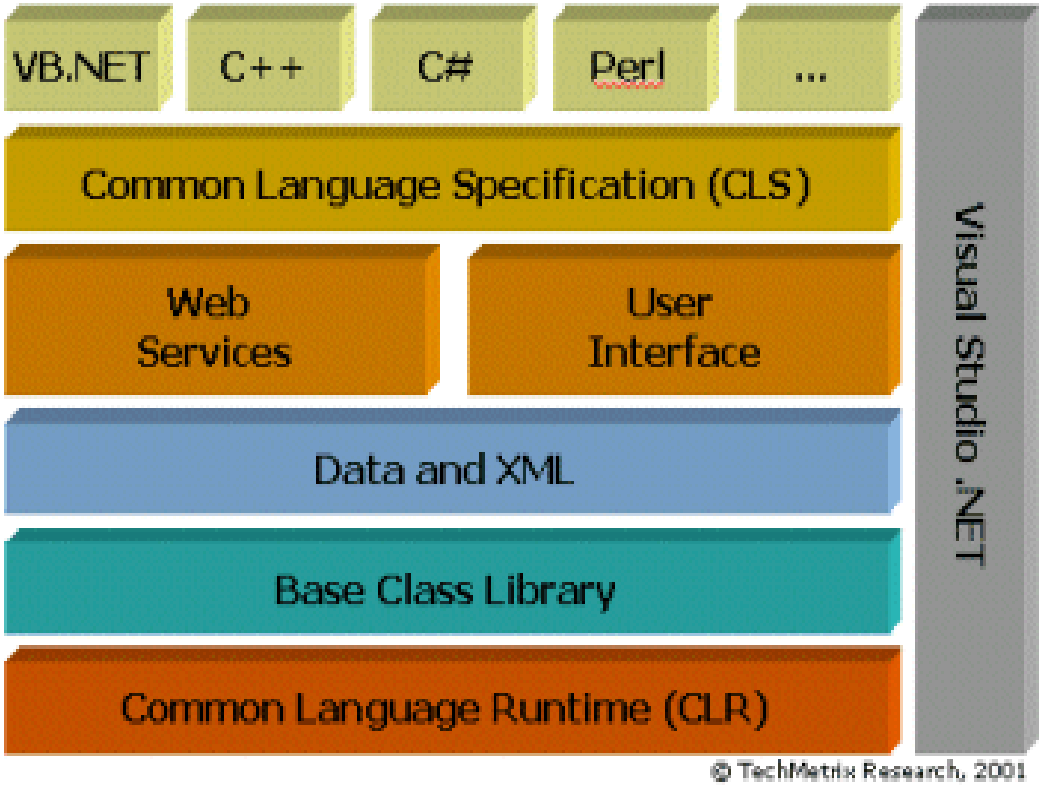
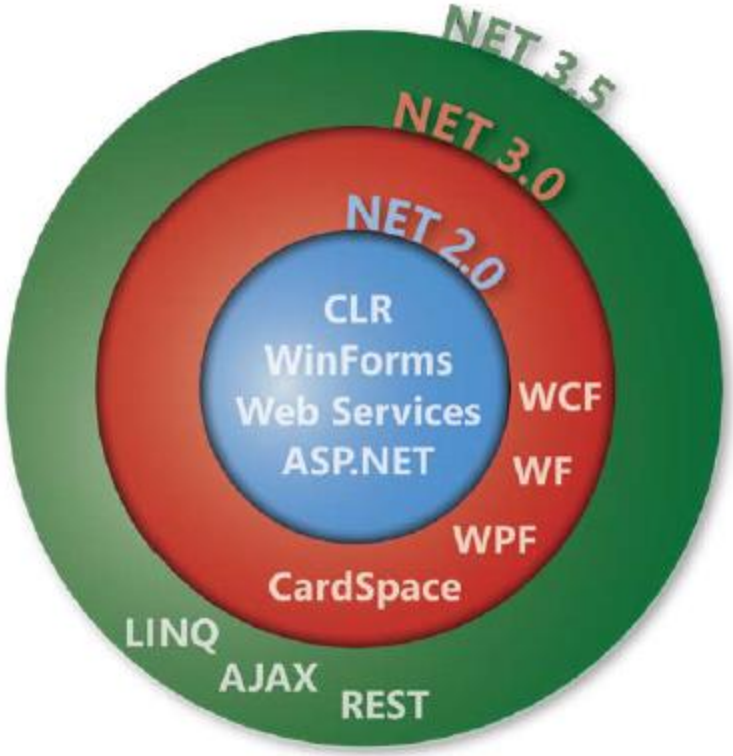
What is .Net Framework?

- .Net Framework is a software development platform developed by Microsoft for building and running Windows applications.
- The first version was released in the year 2002, it was called .Net framework 1.0.
- It consists of:
  - developer tools,
  - programming languages such as Visual Basic and C#, and
  - libraries
- To build desktop and web applications
- It is also used to build websites, web services, and games

Microsoft's .NET Technologies

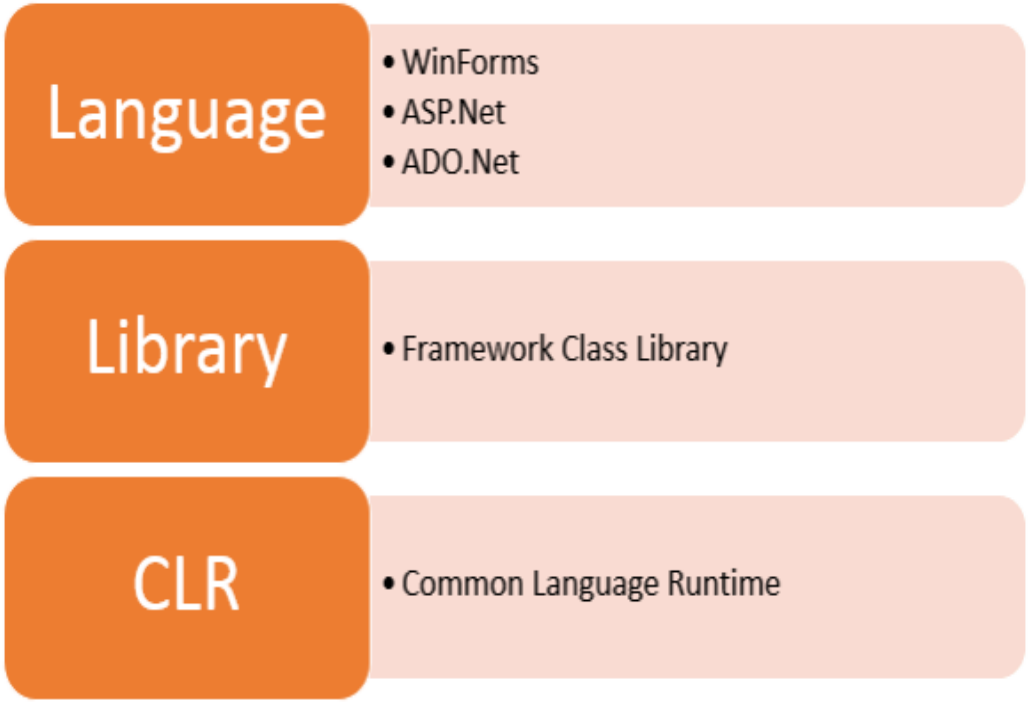




Additive versions of the .NET Framework

.Net Framework Architecture

- A programming model for the .Net platform that provides an execution environment and integration with various programming languages for simple development and deployment of various Windows and desktop applications.
- It consists of class libraries and reusable components.



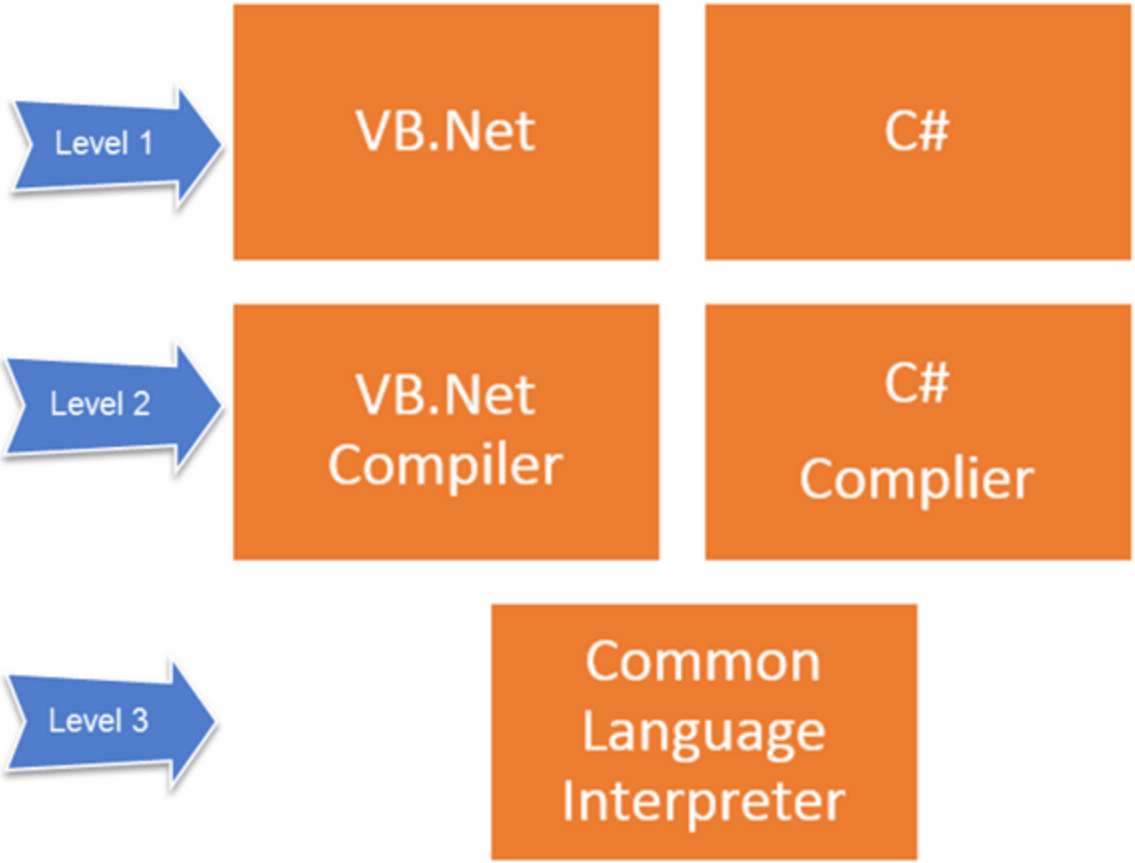
.Net Framework Architecture Diagram

.NET Components
1. Common Language Runtime

- The “Common Language Infrastructure” or CLI is a platform in .Net architecture on which the .Net programs are executed.
- The CLI has the following key features:
  - Exception Handling – Exceptions are errors which occur when the application is executed.
    - If an application tries to open a file on the local machine, but the file is not present.
  - Garbage Collection – Garbage collection is the process of removing unwanted resources when they are no longer required.
    - The database connection is no longer required. If the application has finished all operations on a database, then the database connection may no longer be required.

Working with Various programming languages

1. Language – The first level is the programming language itself, the most common ones are VB.Net and C#.
2. Compiler – There is a compiler which will be separate for each programming language.



3. Common Language Interpreter – This is the final layer in .Net which would be used to run a .net program developed in any programming language. So the subsequent compiler will send the program to the CLI layer to run the .Net application.

2. Class Library

- The .NET Framework includes a set of standard class libraries.
- A class library is a collection of methods and functions that can be used for the core purpose.
  - For example, there is a class library with methods to handle all file-level operations.
  - So there is a method which can be used to read the text from a file. Similarly, there is a method to write text to a file.
- Most of the methods are split into either the System.* or Microsoft * namespaces.
- A namespace is a logical separation of methods.

3. Languages

The types of applications that can be built in the .Net framework:

- WinForms

  This is used for developing Forms-based applications, which would run on an end user machine.

  - Notepad is an example of a client-based application.

- ASP.Net

  This is used for developing web-based applications, which are made to run on any browser such as Chrome.

  The Web application would be processed on a server. The result of the execution is then sent to the client machines, and the output is shown in the browser.

- ADO.Net

  This technology is used to develop applications to interact with Databases such as Oracle or Microsoft SQL Server.

.Net Framework Design Principle

1) Interoperability

An application built on an older version of the .Net framework, would still work on a machine which had the higher version of the .Net framework.

2) Portability

Applications built on the .Net framework can be made to work on any Windows platform.

3) Security

The .NET Framework has a good security mechanism.

- Every application can explicitly define their security mechanism.
- Each security mechanism is used to grant the user access to the code or to the running program.

.Net Framework Design Principle…

4) Memory management

The Common Language runtime does all the work or memory management.

"Garbage Collector" - runs at regular intervals and keeps on checking which system resources are not utilized, and frees them accordingly.

5) Simplified deployment

The .Net framework also have tools, which can be used to package applications built on the .Net framework.

Visual Studio

- Microsoft's Integrated Development Environment (IDE) for creating, running and debugging any application written in C# or any other .NET programming languages.

- When you open Visual Studio it initially displays the **Start Page** (Fig. 2.1).
- Depending on your version of Visual Studio, your Start Page may look different.
- The Start Page contains a list of links to Visual Studio resources and web-based resources.
- At any time, you can return to the **Start Page** by selecting **View > Start Page**.

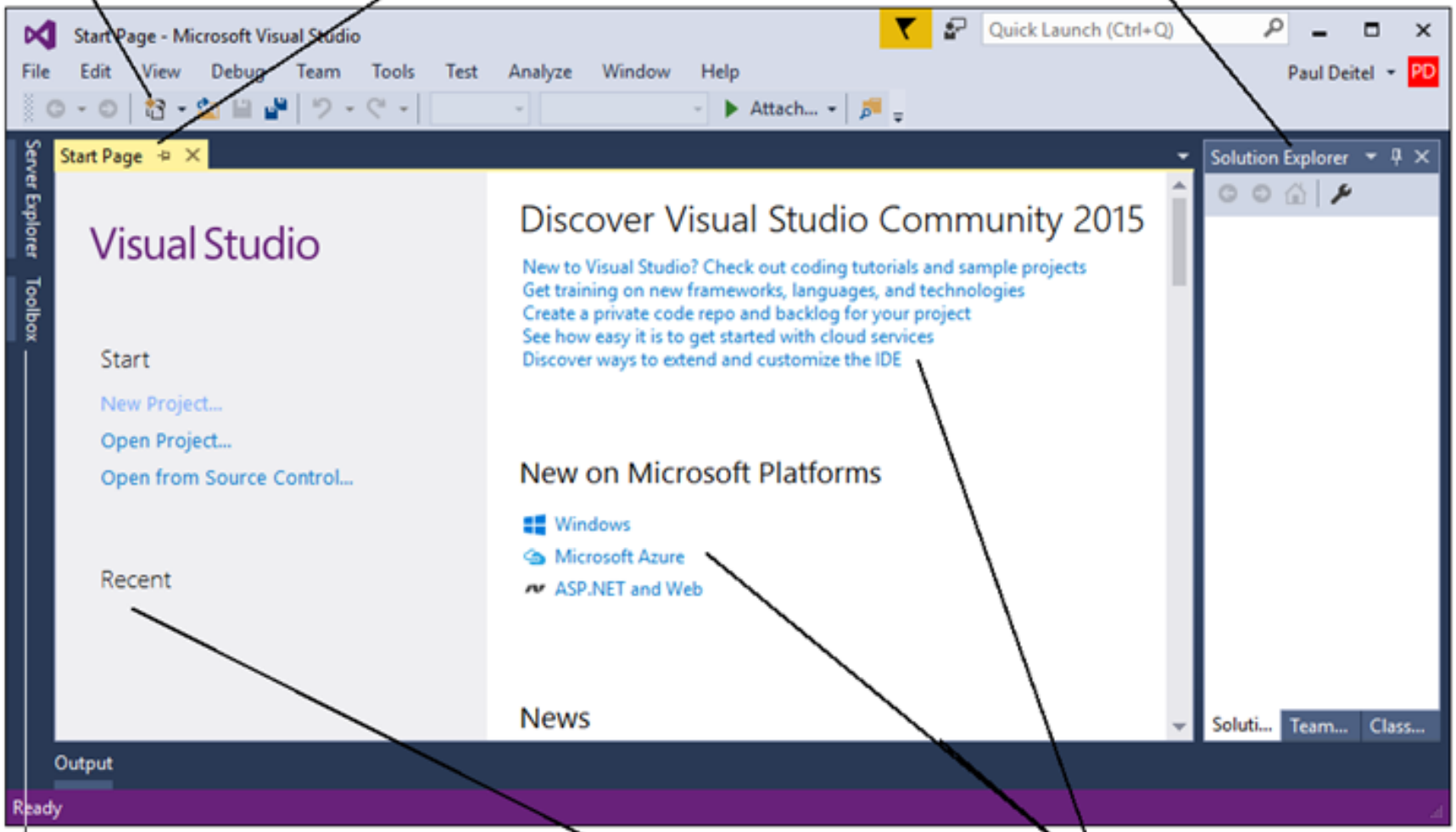


Fig. 2.1 | **Start Page** in Visual Studio Community 2015.

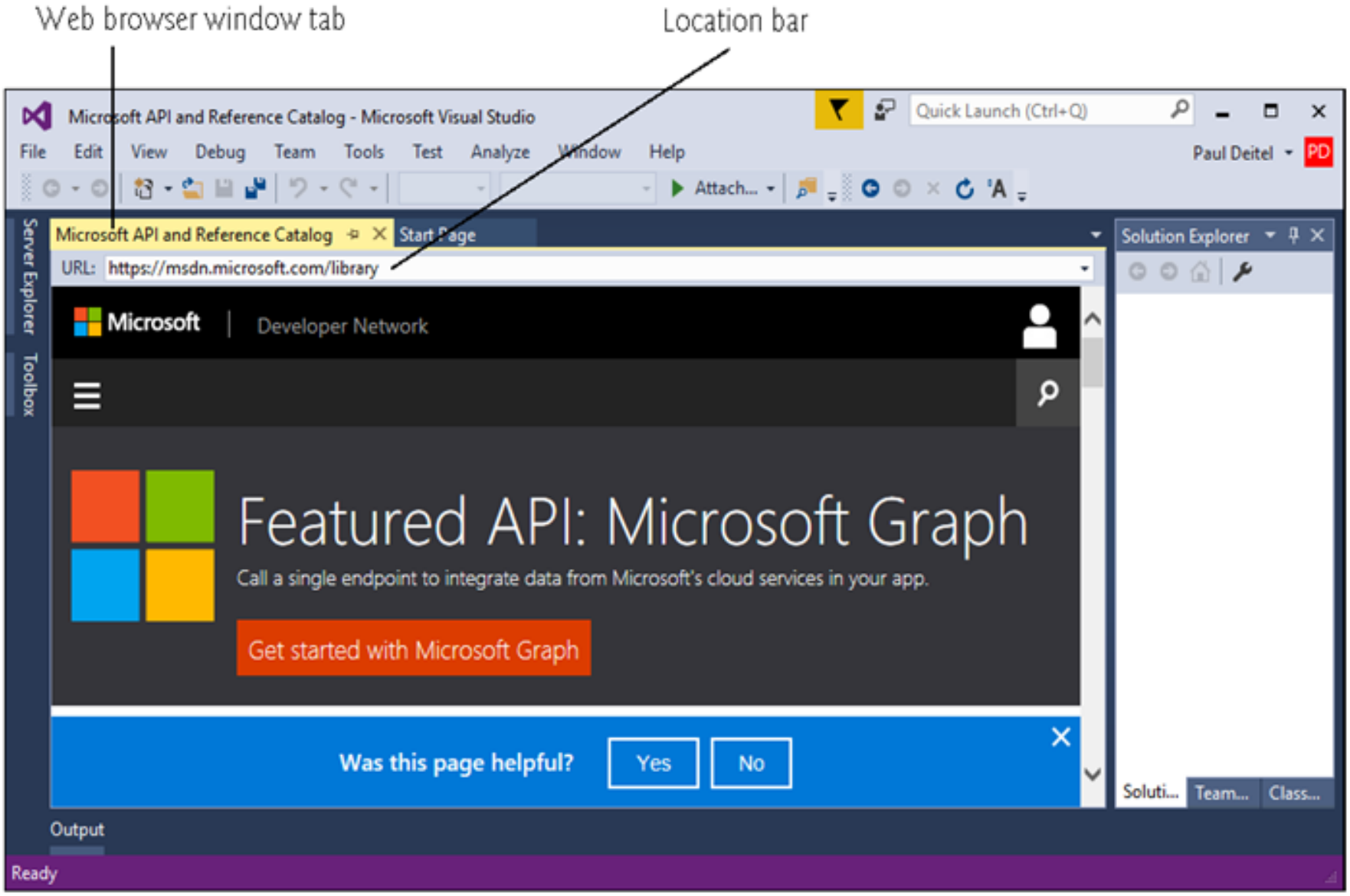


Fig. 2.2 | MSDN Library web page in Visual Studio.

Creating a New Project

- A **project** is a group of related files, such as the Visual C# code and any images that might make up an app.
- Visual Studio organizes apps into projects and **solutions**, which contain one or more projects.
  - Multiple-project solutions are used to create large-scale apps.
- You select **File > New > Project...** to create a new project (Fig. 2.3) or **File > Open > Project/Solution...** to open an existing one.
- You also can click the corresponding links in the **Start Page**'s **Start** section.

- **Dialogs** are windows that facilitate user–computer communication.

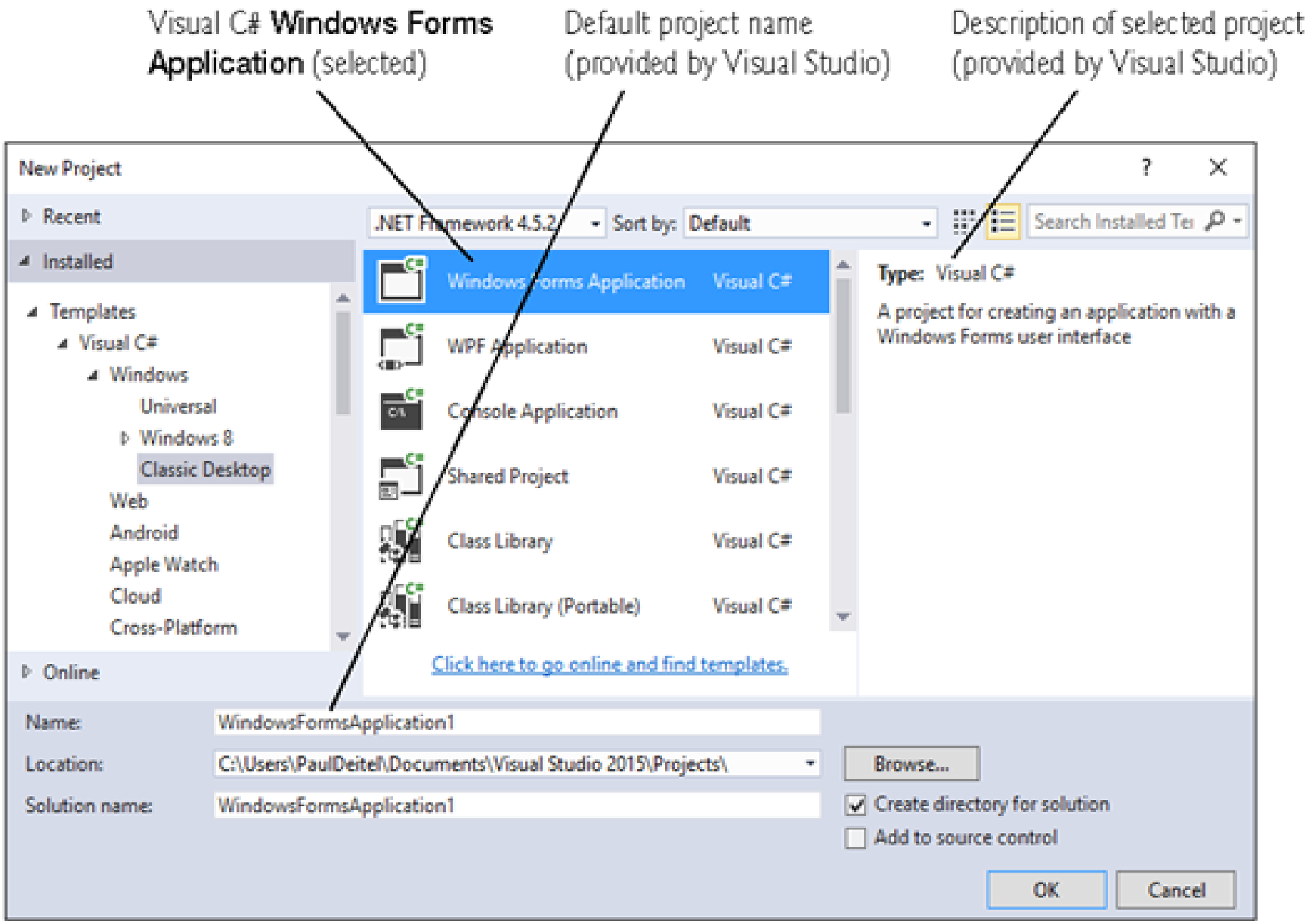
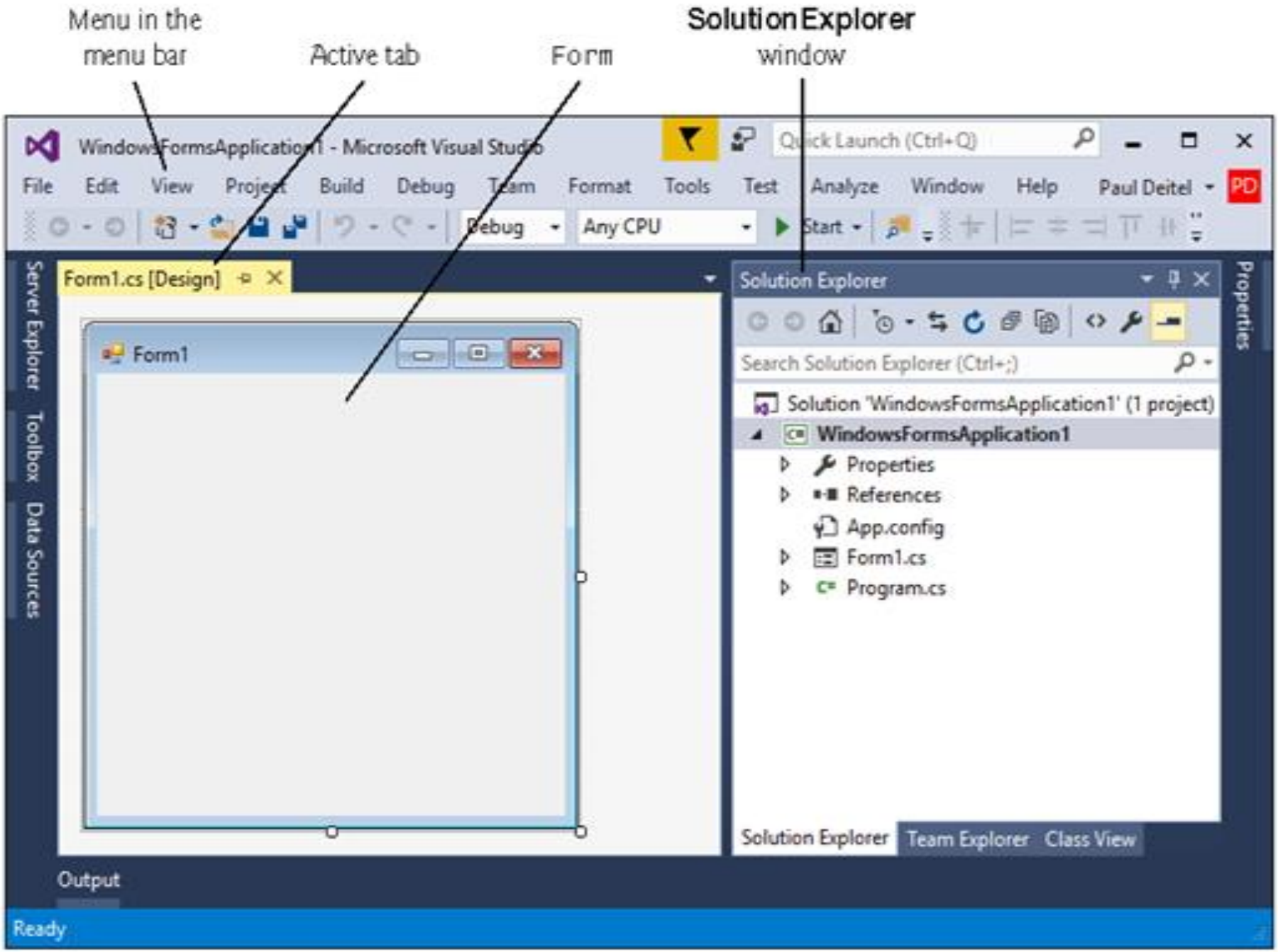


Fig. 2.3 | **New Project** dialog.

New Project Dialog and Project Templates…

- Visual Studio provides many **templates**
  - the *project types* that users can create in Visual C# and other languages.
- A **Windows Forms Application**
  - an app that executes within a Windows operating system and typically has a **graphical user interface** (**GUI**).
    - Users interact with this *visual* part of the app.
- To create a **Windows Forms Application**, under **Templates**
  - select **Visual C# > Windows > Classic Desktop**, then in the middle column select **Windows Forms Application**.
- Click **OK** to display the IDE in **Design view** (Fig. 2.4), which contains the features that enable you to create an app's GUI.


Menu in the menu bar
Active tab
Form
Solution Explorer window
WindowsFormsApplication1 - Microsoft Visual Studio
Quick Launch (Ctrl+Q)
File
Edit
View
Project
Build
Debug
Team
Format
Tools
Test
Analyze
Window
Help
Paul Deitel
PD
Debug
Any CPU
Start
Server Explorer
Toolbox
Data Sources
Form1.cs [Design]
Form1
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'WindowsFormsApplication1' (1 project)
WindowsFormsApplication1
Properties
References
App.config
Form1.cs
Program.cs
Properties
Solution Explorer
Team Explorer
Class View
Output
Ready